改变世界的科学巨匠

The Scientific Giant Who Changed the World



爱因斯坦因为什么获得诺贝尔奖?

READ 纯净阅读

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RECALL 回想巩固

英文原文

On 29 May 1919, as a total eclipse obscured the sun, two British expeditions photographed stars in its apparent vicinity. Arthur Eddington observed from Príncipe, an island off West Africa; Crommelin and Davidson from Sobral, in north-eastern Brazil. They were testing general relativity's bold prediction that the sun's gravity would deflect starlight.

On 6 November, at a joint meeting of the Royal Society and the Royal Astronomical Society, the findings were hailed as confirming it. The Times proclaimed a "Revolution in Science"; a New York Times headline read "Lights All Askew in the Heavens". Almost overnight, Einstein, eminent among physicists, became a global celebrity.

Stockholm proved harder to dazzle. The Nobel physics committee valued rigorous experimental confirmation, and committee member Allvar Gullstrand, a Nobel-winning eye specialist, deemed the empirical support for relativity insufficient. Since no nomination fulfilled Nobel's criteria, the 1921 prize was reserved for a year, as the statutes permit.

On 9 November 1922 the Royal Swedish Academy of Sciences conferred it on Einstein, citing his services to theoretical physics, especially his discovery of the law of the photoelectric effect. That law, which governs how light ejects electrons from metal, helped underpin quantum theory. A letter the next day clarified that the prize took no account of relativity. Einstein, then aboard a steamer bound for Japan, would miss the December ceremony.

On 11 July 1923, in Gothenburg, Sweden, he delivered his Nobel lecture before a congress of Nordic scientists, devoting it to the fundamental ideas and problems of relativity. The theory the committee looked past reached a Nobel podium after all, carried there by the man the prize had honoured.

中文译文

1919年5月29日，日全食遮住太阳时，两支英国考察队拍摄了天空中看似靠近太阳的恒星。阿瑟·爱丁顿在西非外海的普林西比岛观测，克罗姆林和戴维森则在巴西东北部的索布拉尔观测。他们要检验广义相对论的一个大胆预言：太阳的引力会让星光发生偏折。

同年11月6日，在英国皇家学会和皇家天文学会的联席会议上，这些观测结果被盛赞为证实了该预言。《泰晤士报》宣告“科学革命”；《纽约时报》则以“天上的光全歪了”为标题。几乎一夜之间，原本已在物理学界享有盛名的爱因斯坦，变成了全球名人。

斯德哥尔摩的诺奖评审却没那么容易被打动。诺贝尔物理学委员会重视严格的实验证实，而委员阿尔瓦·古尔斯特兰德——一位获得过诺奖的眼科专家——认为相对论的实证支持不足。由于没有一项提名符合诺贝尔规定的标准，1921年的物理学奖按照章程保留一年。

1922年11月9日，瑞典皇家科学院把这个奖授予爱因斯坦，表彰他对理论物理学的贡献，尤其是发现光电效应定律。这一定律揭示光如何使金属释放电子，并帮助奠定了量子理论的基础。授奖决定次日的一封信明确说明，奖项并未考虑相对论。爱因斯坦当时乘轮船前往日本，因此未能参加12月的颁奖典礼。

1923年7月11日，爱因斯坦在瑞典哥德堡的北欧科学家大会上发表诺贝尔演讲，专讲相对论的基本思想与问题。那个被委员会绕开的理论，终究登上了诺贝尔讲台；把它带上去的，正是获得这个奖的人。

核心词汇

先记这22项。连着下方的原文短语一起回想。

eclipse n.
日食；月食
as a total eclipse obscured the sun,

obscure v.
遮住；使模糊
as a total eclipse obscured the sun,

expedition n.
考察队；远征
British expeditions photographed stars

vicinity n.
附近
stars in its apparent vicinity.

deflect v.
使偏折
sun's gravity would deflect starlight.

findings n.
研究或观测结果
the findings were hailed as confirming

hail as v.phr.
被盛赞为
the findings were hailed as confirming

eminent adj.
卓越的；著名的
Almost overnight, Einstein, eminent

rigorous adj.
严格缜密的
rigorous experimental confirmation,

confirmation n.
证实
rigorous experimental confirmation,

deem v.
认定；认为
deemed the empirical support for

empirical adj.
实证的；以观察实验为依据的
deemed the empirical support for

insufficient adj.
不足的
support for relativity insufficient.

criterion n.
标准
nomination fulfilled Nobel's criteria,

reserve v.
保留；暂缓颁发
the 1921 prize was reserved for a

confer on phr.v.
与on连用：授予
Swedish Academy of Sciences conferred

underpin v.
为……奠定基础
helped underpin quantum theory.

clarify v.
澄清；明确说明
letter the next day clarified that the

take no account of v.phr.
不予考虑
prize took no account of relativity.

bound for adj.phr.
驶往；开往
then aboard a steamer bound for Japan,

fundamental adj.
基本的；根本的
to the fundamental ideas and problems

honour v.
表彰；授予荣誉
by the man the prize had honoured.

文中词汇与搭配

补充词汇共33项。保留每个表达在本文中的意思。

observe v.
观测

general relativity n.phr.
广义相对论

bold adj.
大胆的

gravity n.
引力

joint adj.
联合的；联席的

proclaim v.
宣告

headline n.
新闻标题

overnight adv.
一夜之间

physicist n.
物理学家

celebrity n.
名人

dazzle v.
使赞叹；使目眩

experimental adj.
实验的

committee n.
委员会

specialist n.
专家

nomination n.
提名

fulfil v.
符合；满足

statute n.
章程；成文规定

permit v.
允许

on prep.
标示获授予的人

cite v.
列举；引作表彰理由

govern v.
支配；决定

eject v.
使喷出；释放出

electron n.
电子

quantum adj.
量子的

aboard prep.
在船上；乘船

steamer n.
轮船

ceremony n.
典礼

deliver v.
发表演讲

lecture n.
学术演讲；讲座

congress n.
大会

devote to phr.v.
与to连用：专门用于

to prep.
标示投入的对象

podium n.
讲台

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 the sun's gravity would _____ starlight.

02 valued _____ experimental confirmation,

03 helped _____ quantum theory.

04 the prize _____ relativity.

文章主线

- 成名 1919年日食观测被誉为证实太阳引力会使星光偏折，这一相对论预言让爱因斯坦迅速举世闻名。
- 分歧 委员古尔斯特兰德认为相对论证据不足；1921年因无提名达标，物理学奖按章程保留一年。
- 授奖 1922年，保留的1921年度物理学奖授予爱因斯坦，表彰理论物理贡献，尤其是发现光电效应定律。
- 区别 光电效应指光使金属释放电子，这一定律帮助奠定量子理论基础；授奖决定次日的信件明确：该奖不考虑相对论。
- 演讲 爱因斯坦赴日错过颁奖典礼；1923年，他的诺贝尔演讲却专讲相对论，让被奖项绕开的理论登上讲台。

参考答案　1 deflect 2 rigorous 3 underpin 4 took no account of